One Degree of Freedom Moves the Frontier, the Other Four Do Not: A Preregistered Factorization of an Active-Dendrite Repair, Its Architecture-Bound Supply, and Five Failed Escapes

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Abstract

A companion audit established that a one-parameter bias repair to an active-dendrite continual learner acts as a structured-capacity intervention, and conjectured that its gain–retention frontier could only be moved by changing *what* protection covers rather than *when* it applies. We test that conjecture and four adjacent escapes in a series of six preregistered experiments (1,539 newly collected training runs plus 60 reused controls, every criterion and Holm family fixed before data collection, every deviation logged in an unmodified preregistration). The five escapes of the title are the four failed freedoms — *when*, λ, the forward intervention, and the context pathway — plus a fifth that fails loudest of all: the transfer of the one freedom that works. **The conjecture holds on its home architecture.** Reallocating the anchor budget across past tasks under exact total-weight conservation — protecting recent tasks more, old tasks less — moves the frontier where two preregistered decay schedules and every scalar edit failed: at T=20 it adds $\Delta\ell = +8.08$ over the un-repaired baseline while *improving* retention versus the repair itself (ΔBWT $= +1.52$ vs. the repair, Holm $p = 6.4 \times 10^{-5}$; four inherited success criteria met in full), and the direction is specific — protecting *old* tasks instead is worse on retention in all 20/20 paired seeds. **But the supply of this degree of freedom is architecture-bound, and we map its boundary three ways.** On a calibrated real backbone (ResNet-18+GN, λ re-calibrated to 0.1) the same reallocation does nothing (n=20: $\Delta\ell = +0.72$, $p = 0.34$, seed directions 11 positive / 9 negative) — after a first round at n=10 had *appeared* to transfer (+1.94, Holm $p = 0.041$), a preregistered extension clause exposed the small-sample artifact, a flip we report in full. On the convolutional toy backbone, sweeping λ across four decades shows the companion audit's "repair collapses on conv" was partly a single-λ artifact (λ=1000: +0.19, n.s.; $\lambda \leq 316$: +0.80 to +1.04, Holm-significant in 4/6 points) yet the restored effect remains an order of magnitude below its MLP value. A preregistered Transformer extension completes the transfer map at a third trunk family: with both architecture constants re-measured from scratch (G frozen by probe, λ^*=100 from a 120-run sweep), the reallocation is again inert (n=40: $\Delta\ell = +0.09$, n.s. at a detectable threshold of $\approx$0.2 pp — a measured zero, not a missed effect) while the gate normalization itself transfers cleanly (+7.01, 40/40 paired seeds positive): the two freedoms the MLP conflates — normalization and direction — separate across architectures. Finally, training-time corruption of the task context itself — pushed through forward pass, gradient mask, and anchor statistics simultaneously — leaves final accuracy untouched (24 Holm-corrected cells — two consumption modes $\times$ three corruption strengths $\times$ two dose levels $\times$ two datasets; minimum $p = 0.585$), extending a prior evaluation-time null to training, and we show why this null admits a trivial competing explanation (a constant gate that cannot carry information) that the audit's three-instrument protocol can localize. We distill the process into eight preregistration-gate failures we caught by fixed discipline rather than post-hoc judgment, each documented with the diagnostic that distinguished implementation

error from clause conflict. The pattern across all six experiments is a repair with exactly one live degree of freedom whose reach is set by the trunk it sits on.

1 Introduction

Active-dendrite continual learners interleave two mechanisms: a dendritic segment competition that routes each input, and usage-localized consolidation (DeLP) that protects past work by weighting per-neuron Fisher anchors with segment usage. A companion audit (Anonymous, 2026b;a) established, across ten preregistered findings, that the family's one-parameter bias repair is best understood as a *structured-capacity* intervention: its learning-term gain is reproduced by scaling the protection strength λ alone, its retention cost concentrates on recent tasks, and changing *when* the intervention applies — two preregistered decay schedules — does nothing to separate gain from forgetting. The audit ended with a conjecture, stated as the natural next experiment: that the gain–retention frontier can only be moved by changing *what* the protection covers.

This paper is the systematic test of that conjecture and of every adjacent escape we could preregister. It comprises six experiments, R1–R4 plus a transfer extension R2T — 1,539 newly collected training runs and 60 reused audit controls, across two datasets, five architectures, and a real backbone. Every experiment was run under a written preregistration whose success criteria, failure criteria, statistical families, and conditional triggers were fixed before any data was collected; where a gate failed or a clause had to be revised, the preregistration was left unmodified and the deviation was recorded in a separate revision note. Section 9 summarizes this process because we believe it is itself a contribution: eight implementation-gate failures, each caught by a deterministic check rather than judgment, each with a different root cause.

The five questions. Let $a_1, \ldots, a_v$ be the Fisher anchors accumulated after v tasks, each with weight implicitly 1. The repair (retaining normalization of the gated trunk) sits on top of these anchors. The degrees of freedom available to a practitioner are: (i) *what* the budget protects — the per-anchor weights w_j; (ii) *when* it applies — the audit's decay schedules, already falsified; (iii) *how strongly* — λ, shown by the audit to be the dominant scalar axis; (iv) the *forward* intervention itself; and (v) the *context pathway* that supposedly routes tasks. We preregister one experiment per remaining freedom and report all of them regardless of outcome.

Findings. **F1 (what):** budget direction is live on the home architecture — weight conservation notwithstanding, shifting weight toward recent tasks under $\sum_j w_j = v$ moves the frontier (four inherited criteria, Holm $p = 6.4 \times 10^{-5}$), and the direction is causal (20/20 paired seeds reverse when the weights point at old tasks). Specificity controls show the effect requires the repaired forward pass: reallocation without it improves retention but produces no gain, and raising λ alone slides *along* the frontier rather than moving it. **F2 (transfer):** the same intervention is inert on a calibrated ResNet-18+GN backbone at n=20 — and we report the n=10 round that first appeared positive and was overturned by the preregistered extension, as a concrete instance of a discipline preventing a false positive — and inert again on a Transformer at n=40 with both architecture constants recalibrated from scratch, while the normalization itself transfers there cleanly: the supply map gains a third point, and the two freedoms separate. **F3 (λ attribution):** the companion audit's conv collapse was partly a single-point artifact; sweeping λ restores a significant but order-of-magnitude-smaller effect, with the audit's chosen λ sitting exactly at the collapse point. **F4 (context):** training-time task-label corruption through all three consumption sites leaves accuracy untouched, with a structural competing explanation we make explicit rather than claim robustness. **F5 (process):** eight gate failures, all caught by fixed checks; we extract portable rules.

Contributions. (1) A preregistered confirmation, with full specificity controls, that anchor-budget *direction* is the second live degree of freedom of this repair family, and that it is the only one besides λ. (2) A four-architecture boundary map of where that freedom runs out: real backbone (no effect at n=20), miscalibrated toy conv (effect masked at the audited λ, restored but 19% strength elsewhere), Transformer (no effect at n=40 with recalibration, while normalization transfers at 40/40 seeds), and the context pathway (structurally unable to carry the signal). (3) A methods section documenting eight preregistration-gate

Table 1: Training protocol, inherited verbatim from the companion audit per backbone. One row per hyperparameter; nothing in any column was tuned by this paper. MLP and conv share the audit's full training stack (optimiser, schedule, sample budget); ResNet-18+GN carries the audit's external-validity calibration, in which the only re-calibrated quantity is λ (and the repair's G); the Transformer (R4) re-measures both constants from scratch by preregistered probe and sweep.

	bare MLP (R1, R2)	toy conv (R3)	ResNet-18+GN (R2T)	Transformer (R4)
datasets	Permuted-MNIST / Permuted-Fashion	Permuted-MNIST / Permuted-Fashion	Split-CIFAR-100 (task-IL)	Permuted-MNIST / Permuted-Fashion
tasks T	10, 20	10	10	20
trunk	MLP, width 256 one segment/task	4-layer conv, bias-free, width 2048	ResNet-18+GN, pooled width 512	2-layer Transf., d=256, pooled 512
segment keep (kWTA)	25% (64/256)	25%	25% (128/512)	25% (128/512)
optimiser	Adam, lr=10^{-3}	Adam, lr=10^{-3}	SGD, lr=5×10^{-2}	Adam, lr=10^{-3}
momentum / weight decay	–	–	0.9 / 5×10^{-4}	–
gradient clip	–	–	global norm 5.0	–
batch / epochs per task	256 / 2	256 / 2	128 / 5	256 / 2
train samples per task	10,000	10,000	5,000 (10 classes)	10,000
λ (audited)	1000	1000	0.1	100 (sweep)
gate constant G	0.565	0.565	0.5749	0.5827 (probe)

Task identity is seeded per seed (generator seed 100+seed): within a seed, every arm sees identical task permutations (MLP/conv/Transformer) or identical class orders (ResNet), which is what makes the paired comparisons exact. The conv trunk is 32→64→128→2048 channels, 3×3, stride 1/2/2/2, no bias anywhere. G is the measured constant gate expectation, per backbone; the repair divides the trunk by it. The R4 column's λ and G are this paper's own calibration outputs (Section 8), not inherited values.

failures and their diagnoses, which we argue should accompany any paper claiming preregistered discipline. (4) A fully scripted reproduction package (independent recomputation of every headline number from raw run records, 25/25 checks passing on two machines).

2 Background and Setting

We inherit the companion audit's protocol in full; we restate only what the new experiments manipulate. The learner is the bare-MLP active-dendrite network of the audit (width 256, n_{seg}=T, kWTA keep-rate 25%, protected gradients on), trained on Permuted-MNIST and Permuted-Fashion with $T \in \{10, 20\}$ tasks, Adam, oracle task context at evaluation. Metrics are the learning term ℓ (mean just-learned accuracy), final average $\bar{a}$, and backward transfer BWT; all comparisons are paired by (dataset, seed) with n=10 per dataset, n=20 pooled, two-tailed, Holm-corrected within families fixed in the preregistrations.

The repair under study deletes the sigmoid gate and rescales the trunk by $1/G$ with G the measured constant gate expectation (Anonymous, 2026b;a); we call this arm `renorm`. Its anchors are the per-neuron Fisher anchors of DeLP, accumulated one per completed task with unit weight. The audit's four success criteria for any candidate frontier-mover at T=20: $\Delta\ell \geq +4.0$ over the un-repaired baseline, $\Delta\text{BWT} \geq -0.5$, improvement of $\Delta\text{BWT} \geq +1.11$ over `renorm` itself with Holm $p < 0.05$, and no degradation at T=10. These are inherited verbatim; we add nothing.

3 Related Work

Regularization-based continual learning. The parameter-protection literature splits into two families. The regularization family (Kirkpatrick et al., 2017; Zenke et al., 2017; Aljundi et al., 2018; Chaudhry et al., 2018) concentrates its effort on one question: how should *per-parameter* importance be estimated — Fisher information, path-integral accumulation, sensitivity to perturbation — and imposed; the isolation family masks or prunes task-specific parameter subsets (Serra et al., 2018; Mallya & Lazebnik, 2018). In both, the estimated protection is a measurement; the research variable is the estimator or the mask. This paper holds the estimator fixed (the audit's per-neuron Fisher anchors, inherited verbatim) and varies a question upstream of estimation that neither family's design space otherwise exposes: how a *per-anchor, task-level*

protection budget is *allocated* across past tasks, under exact total-weight conservation that removes regularization mass as a confound. On that axis the families' own knobs are scalar-, schedule-, or mask-shaped — a global λ (Kirkpatrick et al., 2017), online decay of consolidation strength (Chaudhry et al., 2018), hard attention masks (Serra et al., 2018) — which is precisely the *strength/schedule* axis the companion audit found exhausted and this paper's R3 re-quantifies. Context-dependent gating, the mechanism whose repair created these anchors, has its own lineage (Masse et al., 2018); R1 tests whether its context pathway matters in training. Surveys delimit the same axes (De Lange et al., 2022; Parisi et al., 2019); to our knowledge no prior continual-learning work isolates a direction-of-allocation effect under exact budget conservation.

Dendritic computation. The structural substrate — compartmentalized dendrites computing context-dependent nonlinearity (Poirazi & Papoutsi, 2020; Ujfalussy et al., 2018), consolidation dynamics (Benna & Fusi, 2016), and dendritic gating of plasticity (Onasch et al., 2026) — motivates treating the anchor population as structured capacity rather than a scalar regularizer; the companion audit's gate-constant finding and this paper's direction specificity are the quantitative expression of that framing.

Methodology. The discipline apparatus follows the preregistration and honest-reporting line (Lipton & Steinhardt, 2019; Musgrave et al., 2020; Agarwal et al., 2021): criteria and Holm families fixed before data collection, sample-extension and reverse-trigger clauses fixed in advance, every deviation logged against an unmodified preregistration, and every number in this paper mechanically recomputed from raw records.

4 R2: The Anchor-Budget Direction

4.1 Intervention

At phase v with anchors $a_0, \ldots, a_{v-1}$ of ages $g_j = (v-1) - j$, replace the unit weights by

$$w_j \;=\; (1-\alpha) \;+\; \alpha \,\frac{v\, e^{-g_j/\tau}}{\sum_k e^{-g_k/\tau}}, \qquad \tau = T/4, \tag{1}$$

so that $\sum_j w_j = v$ identically for every α: total protection strength is conserved phase by phase, and α=0 reproduces the audit's uniform penalty bit-for-bit (verified by an independent sentinel grid, 40/40 pairs, maximum metric difference 0.0). This conservation is the design feature that isolates *direction* from *strength* — the audit had already shown λ (strength) to be the dominant scalar axis, and any "direction" result that quietly changed strength would be uninterpretable. Nothing else changes: not the forward pass, not anchor construction, not Fisher estimation, not evaluation.

4.2 Preregistered criteria and arms

The six arms are `base`, `renorm`, the candidates `renorm_rec05` (α=0.5) and `renorm_rec10` (α=1), the specificity control `renorm_lamup` ($\lambda\times$3.13 on `renorm`), and the interaction control `base_rec10` (reallocation without the repaired forward pass), over 2 datasets $\times\, T \in \{10, 20\} \times 10$ seeds (240 runs). The Holm family is the two candidates' ΔBWT versus `renorm` at T=20, fixed in advance.

4.3 Result: the frontier moves

Table 2: `renorm_rec10` meets all four inherited criteria; $\Delta\ell$ seed directions are 20/20; we report honestly that ΔBWT-versus-`renorm` directions are 18/20, short of the strict unanimity form, and we do not downgrade the criterion to pass it. The preregistered reverse-weighting trigger fired: protecting *old* tasks (`renorm_anti10`) yields *higher* learning ($\Delta\ell = +11.62$) but collapses retention (ΔBWT $= -6.68$; versus `rec10`: -7.05, 20/20 paired seeds, $|d_z| = 4.0$). Direction, not strength or schedule, is the carrier.

Figure 1 places the six arms on the gain–retention plane. Two readings matter. First, `renorm_rec10` dominates `renorm` in both coordinates — this is what "moving" a frontier means, as opposed to sliding along it, and the λ-only control (`renorm_lamup`) traces exactly such a slide. Second, the reverse-weighting arm is *not* a mirror image of the baseline: it buys retention with learning, confirming that the budget is a real

Table 2: R2 headline, pooled over datasets (n=20) with the per-dataset decomposition of the two weakest quantities (learning-term difference versus the repair; n=10 each). Criteria are evaluated on the pooled statistics, inherited verbatim from the companion audit's closure grid; the decomposition is reported, per the preregistration, because pooling masks dataset asymmetry. Holm family = the two candidates' ΔBWT vs. `renorm` (pooled).

	pooled (n=20)						$\Delta\ell$ vs. `renorm`, split	
arm	$\Delta\ell$ (T=20)	ΔBWT (T=20)	ΔBWT vs. `renorm`	$\Delta\ell$ (T=10)	ΔBWT (T=10)	Holm p	MNIST T=20	Fashion T=20
`renorm_rec05`	+7.40	−0.34	+0.81	+5.39	+3.33	1.4×10^{-3}	+0.10	−0.58
`renorm_rec10`	**+8.08**	**+0.37**	**+1.52**	**+5.84**	**+4.45**	$\mathbf{6.4\times10^{-5}}$	+1.10*	−0.22

Split columns: the learning-term difference versus `renorm` within each dataset. *p=0.015; the Fashion split is negative (−0.22, p=0.63) — on Fashion the reallocated repair matches rather than exceeds the repair's learning term, and the pooled ΔBWT-versus-repair improvement is what carries the frontier claim there. Both splits are secondary descriptions; all criteria are pooled, as preregistered.

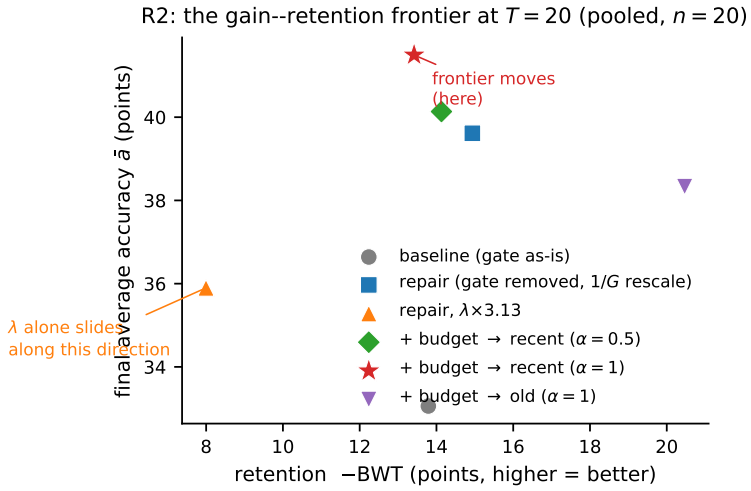


Figure 1: The T=20 gain–retention plane (pooled over both datasets, n=20; horizontal axis is retention −BWT, higher is better). The recency-reallocated repair (`renorm_rec10`, red star) sits above and to the right of the repair itself: the frontier moves. Raising λ alone (orange triangle) moves along the retention–accuracy trade-off, off the frontier. Redirecting the same budget toward *old* tasks (purple) recovers retention but pays 3.2 points of final accuracy: the direction of the budget, not its existence, carries the effect.

instrument with a direction, not a scalar knob in disguise. The shape hyperparameters (τ=$T/4$, $\alpha \in \{0.5, 1\}$) were fixed in the preregistration and never scanned; the reverse-weighting arm uses the identical shape with the sign flipped, so the 20/20 reversal is itself evidence that the finding is not an artifact of one favorable weighting profile — a shape-sensitive artifact would not survive its own mirror.

4.4 Specificity: what the result is *not*

Two controls prevent two over-reads. `renorm_lamup` (ΔBWT vs. `renorm` = +6.94, larger than the candidate's) *loses* 2.66 points of learning: raising strength slides along the frontier (retain more, learn less) — it does not move it, and the preregistered downgrade clause ("if strength-only also qualifies, the claim collapses") does not trigger. `base_rec10` improves retention (+2.25) with *no* gain (−0.44): reallocation alone is a retention instrument; only on the repaired forward pass does it move the frontier. The claim is therefore exactly: *on the repaired baseline, budget direction moves the frontier*; not "recency weighting is a general continual-learning improvement."

5 R2T: The Boundary on a Real Backbone

The companion audit's external-validity arm had already shown the repaired forward pass itself is the *worst* gate manipulation on ResNet-18+GN / Split-CIFAR-100 (−8.83). R2T asks the orthogonal question: does *budget direction* survive where the forward intervention does not? Protocol is the audit's, inherited verbatim by import (λ=0.1, its per-architecture calibration; G=0.5749; SGD; task-IL; 10 seeds): Split-CIFAR-100 as ten tasks of ten classes each, all 5,000 training images per task, and a class order seeded identically across arms so that paired comparisons remain exact (Table 1, right column). New arms are `base_rec10` (reallocation only), `renorm_rec10` (both), and `base_anti10`, against the audit's existing `base`/`renorm` runs reused as shared controls.

The flip. The first round (n=10) supported transfer: $\Delta\ell(\texttt{base_rec10} - \texttt{base}) = +1.94$, Holm $p = 0.041$, apparently qualifying as "reallocation improves the un-repaired backbone too." Three of four primary raw p-values fell in the preregistered $(0.01, 0.1)$ extension band, firing the sample-extension clause. At n=20 — new arms plus control arms rerun for the new seeds with the audit's own trainer — the effect collapsed to +0.72 ($p = 0.34$, seed directions 11 positive / 9 negative); the repaired-contrast collapsed likewise. The preregistered verdict had three levels — T1 transfer, T2 partial rescue (within the repaired arm only), T3 fail — and reads **T3**: the contrast did not even reach the partial-rescue level. We report the n=10 round, its trigger, and the flip in the main text, not an appendix: a discipline that only surfaces its confirmations is advertising, not method. A further reason to distrust the first round operates before any statistics: the R2T pairing is *cross-run* — new arms trained in this paper's sessions, control arms in the audit's — and a design probe run *before* the preregistration, as environment characterization rather than R2T data, measured same-seed reruns of the audit's own baseline arm differing by up to 6.8 learning-term points under cuDNN nondeterminism. That probe is the reason R2T's preregistration contains no bit-identity gates and relies on paired statistics instead. A single-round Holm p=0.041 is exactly the kind of margin that run-level noise can manufacture; the extension clause was designed for marginal significances, and here it doubly earned its keep. Figure 2 shows the per-seed deltas behind the flip: the first round's nine positive seeds were not a stable majority but a lucky draw, and the ten added seeds split 2 positive / 8 negative. The large residual contrast ($\texttt{base_rec10} - \texttt{renorm} = +10.06$, 20/20 seeds) is carried almost entirely by the repaired forward pass being harmful on this backbone — an audit finding, not a new one. Direction specificity is also absent here (reverse weighting is statistically indistinguishable), distinguishing the real backbone from the MLP, where it was 20/20.

6 R3: The λ Attribution on the Toy Conv

The audit's threats-to-validity section conceded the conv collapse might be a λ mis-calibration artifact. R3 sweeps $\lambda \in \{0, 3.2, 32, 316, 1000, 3160\}$ on the audit's conv backbone — the audit's four-layer bias-free conv trunk on the same Permuted-MNIST/Fashion tasks and the same training stack as the MLP (Table 1), with the gate constant inherited unchanged — (240 runs; λ=0, never previously run, included; two arms × two datasets × ten seeds per λ). The preregistered two-sided criteria — "restoration" ($\Delta\ell \geq +2$) versus "boundary" (all $|\Delta\ell| < 1$, no trend) — *both* failed to fire: the truth is in between, and more interesting than either. Learning-term gain over the un-repaired arm is Holm-significant at $\lambda \leq 316$ (+0.80 to +1.04, 4/6 points) yet collapses exactly at the audit's λ=1000 (+0.19, $p = 0.57$), and the restored magnitude is 19% of the MLP value (the audit's +5.48 at the same T, same protocol, same seeds). The curve has a second, shallower dip at λ=3160 (+0.64, Holm $p = 0.070$): the audit's single point sits at the deepest of the two troughs. The audit's collapse was thus partly a single-point artifact — its one chosen λ lands on the collapse region of a non-monotonic dose curve, at its lowest point — but recalibration restores an order-of-magnitude-smaller effect, not the repair. We state the correct claim narrowly: single-λ cross-architecture conclusions can over- *or* under-estimate; here it did both, in different directions for significance and magnitude.

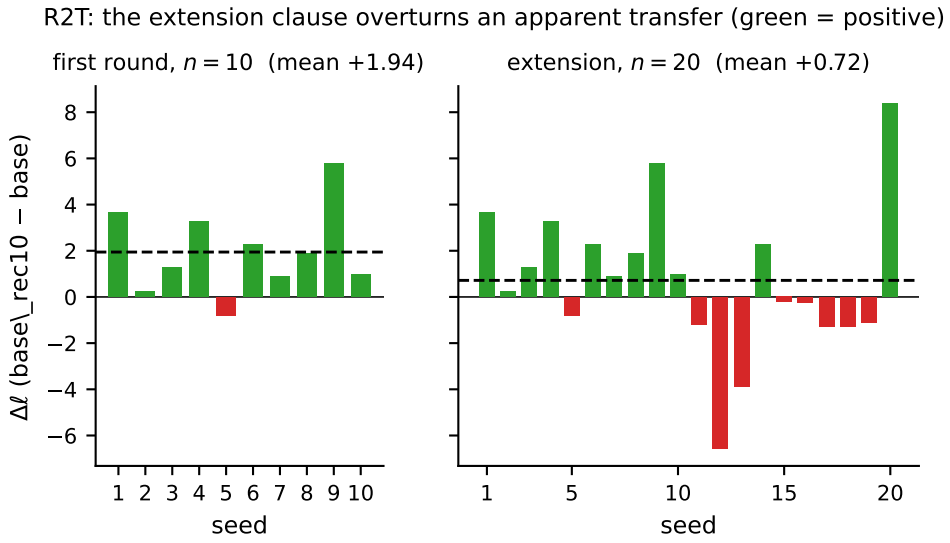


Figure 2: R2T per-seed $\Delta\ell$ (reallocation minus baseline on ResNet-18+GN). Left: the first round's n=10 seeds — mean +1.94, one strongly negative seed, enough for Holm-significance at $p = 0.041$. Right: the preregistered extension to n=20 — the ten added seeds split 2 positive/8 negative, the mean collapses to +0.72 ($p = 0.34$), and the verdict flips from "transfer" to "fail". Green bars are positive seeds.

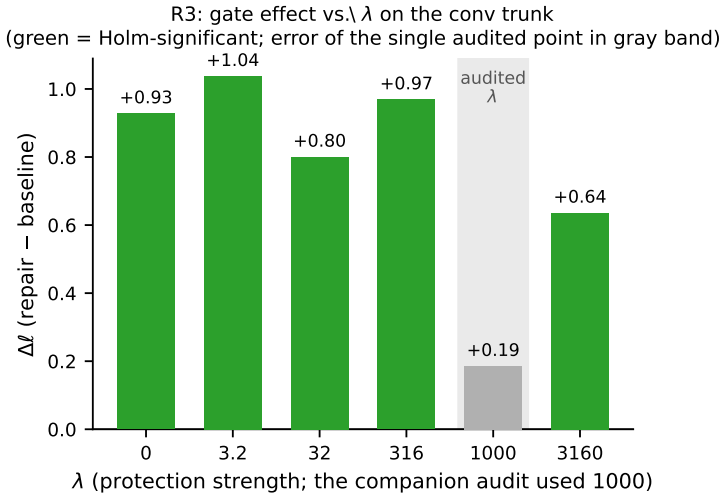


Figure 3: R3 dose curve: learning-term gain of the repair over the un-repaired conv baseline, versus λ (log axis; pooled n=20). Green bars are Holm-significant. The companion audit's single audited point (λ=1000, shaded) sits exactly at the collapse point of the curve — the one value at which the gate effect vanishes — while both weaker protection ($\lambda \leq 316$) and the λ=0 no-anchor control show a significant but order-of-magnitude-smaller effect than on the MLP (+5.48).

7 R1: The Context Pathway in Training

An earlier evaluation-time probe had shown that corrupting the task context at test time barely moves accuracy; the audit flagged training-time corruption as the natural next experiment. R1 runs it: per-phase task labels are mislabeled with probability $\rho \in \{0.1, 0.2, 0.4\}$ (a guaranteed-at-least-one sampling rule, after a preregistered gate caught the Bernoulli version's mathematical conflict with a per-run positivity clause),

simultaneously through all three consumption sites — forward pass, gradient mask, and anchor Fisher/usage statistics — against clean controls and against a *forward-only* arm (corruption at the forward-pass site alone), whose contrast with the full corruption is the preregistered anchor-chain test, 280 authoritative runs. Nothing transfers to accuracy: the smallest Holm-corrected p across all 24 effect cells is 0.585; the preregistered anchor-chain hypothesis (corrupting anchor statistics costs more than corrupting the forward pass alone) fails with $|\Delta| \leq 0.42$; and the repair's bias gain survives corruption at 93–112% of its clean value. We explicitly refuse the robustness reading: the audit's instruments show the gate constant (0.564 across architectures) and the gradient-mask granularity inconsistent with the forward read, so "corrupting a pathway that structurally cannot carry information" is an equally sufficient explanation, and R1 cannot distinguish the two. What R1 does establish, narrowly, is that *training-time label noise need not enter the acceptance checklist* for this family on the audited trunks.

8 R4: The Third Point on the Supply Map

R2T left the supply boundary a two-point line: live on the bare MLP, dead on a real convolutional trunk. A line cannot distinguish "MLP-specific" from "everything except the MLP," so R4 adds a third point from a different family: attention. A two-layer Transformer (d_{model} 256, four heads, 28 row-tokens mean-pooled into a 512-d feature) carries the same dendritic gate on its pooled features — the audit's attachment, with per-parameter Fisher anchors and the same 25% kWTA keep-rate — on the same Permuted-MNIST/Fashion task stream at T=20, with the R2 optimiser protocol inherited verbatim by import (Table 1, right column). Nothing about the intervention is rewritten: the recency weights are imported from the R2 module, and the α=0 identity is asserted constructively on the imported function itself, not re-implemented.

Calibration. Both architecture-relevant constants are re-measured, not inherited. The gate constant is frozen from a base-arm probe ($G = 0.5827$, sd 0.0017 over 120 phase-probes; six runs), and λ is recalibrated by a preregistered 120-run sweep over six decades: the pooled reallocation-favorable gain (renorm − base) rises monotonically from +2.71 at λ=0 to +7.10 at λ^*=100, then eases to +7.00 at 1000 — an inverted U with no second trough, and no all-negative region, so the preregistered fallback to the weaker contrast never fired. The main experiment ran at the calibrated optimum with n=40 (2 datasets $\times$ 20 seeds, paired by (dataset, seed); 200 runs, five arms).

Result: T3 again — and the map now has a trend. All seven implementation gates pass, including a calibration-consistency gate that turns out to be bitwise: the Transformer driver is deterministic on this stack, so scan and main runs agree to the last digit on the overlapping seeds (the cuDNN nondeterminism that R2T measured at 6.8 points on ResNet does not occur here). The preregistered three-level verdict — inherited verbatim from R2T — reads **T3**: reallocation is inert within both bases (Table 3: C1 +0.07, p=0.33; C2 +0.09, p=0.14; the Holm family of four is uniformly ns; seed directions mixed at +26/−14 and +26/−14), and direction specificity is absent for the second real-architecture time (anti vs. rec: indistinguishable). The power note matters as much as the verdict: the achieved paired sd is 0.37–0.45 learning-term points, roughly an order of magnitude below the conservative sd= 3 the design assumed, putting the detectable threshold at ≈ 0.2 pp — the R2 effect (+8.08) would have registered at tens of standard deviations. This is a measured zero, not a missed effect.

What R4 adds: the two freedoms separate. The negative verdict is the least interesting thing in Table 3. The gate normalization itself — the repaired forward pass, which the audit found *harmful* on ResNet (−8.83) — transfers to attention cleanly: renorm − base = +7.01 with all 40/40 paired seeds positive, and the reverse contrast C4 pays back −6.94 with 0/40 sign disagreements. The supply claim therefore splits in two. The *normalization* freedom generalizes across all three tested trunks (MLP +5.48, conv restored to 19% strength by recalibration, Transformer +7.01); the *direction* freedom does not leave the MLP. "Architecture-bound supply" stops being a binary line and becomes a three-architecture trend, and the R2 claim inherits a sharper form: on the MLP, budget direction and normalization were conflated in a single repaired forward pass; the Transformer, where normalization works and direction does nothing, shows they are mechanically independent.

Table 3: R4 headline on the Transformer (T=20, pooled over datasets, n=40 paired by (dataset, seed)). C1/C2 enter the preregistered Holm family (four tests); C3/C4 are exploratory descriptions, as in R2T. Arm means (n=40 each): `base` $\ell = 54.68$ / $\bar{a} = 24.00$ / BWT $= -32.29$; `renorm` 61.69/23.99/−39.69; `base_rec10` 54.75/23.77/−32.60; `renorm_rec10` 61.77/24.05/−39.71; `base_anti10` 54.68/23.79/−32.51.

	comparison	$\Delta\ell$	ΔBWT	$\Delta\ell$ split (MNIST / Fashion)	Holm family
C1	`base_rec10` − `base`	+0.07	−0.31	−0.06 / +0.20	ns
C2	`renorm_rec10` − `renorm`	+0.09	−0.03	+0.08 / +0.09	ns
C3†	`base_anti10` − `base_rec10`	−0.07	+0.09	—	—
C4†	`base_rec10` − `renorm`	−6.94	+7.08	0/40 seeds	—

†Exploratory, outside the family. Holm family raw p: C1 ℓ 0.332, C1 BWT 0.187, C2 ℓ 0.145, C2 BWT 0.904; all Holm-adjusted ≥ 0.58. C4: $p = 1.1 \times 10^{-24}$ (ℓ), 4.4×10^{-16} (BWT), and the ℓ seed direction is 0/40 — the normalization effect is perfectly sign-consistent even where its nominal carrier (reallocation) is inert. Family $|d_z|$: 0.155, 0.213, 0.235, 0.019. C3 seed directions are mixed in every metric — direction specificity is absent on this trunk.

9 The Discipline: Eight Gate Failures

Every experiment above ran under a preregistration with implementation-correctness gates whose failure voids the data. Eight times a gate fired; none was resolved by relaxing a criterion. Table 4 lists them with the diagnosis each demanded — no two share a root cause — because the set illustrates what fixed checks buy that judgment does not.

The portable rules are stated in the reproduction package; we believe items (1), (2), (7) and (8) in particular generalize to any preregistered continual-learning program. A ninth, milder failure occurred at the writing stage and is worth recording in the same spirit: an early draft of Figure 2's caption described the added seeds as "3 positive / 7 negative" and the first round as "eight positive seeds" — both written from the round means rather than counted from the per-seed records (the true values are 2/8 and nine). A scripted check of every count and statistic against the raw run records caught both; the rule we extract is that *caption-level numbers deserve the same programmatic verification as headline statistics*, because they are quoted exactly as often and checked far less.

10 Limitations

All claims are bounded by the audited trunks (bare MLP, GN-MLP, 4-layer conv, ResNet-18+GN, and R4's two-layer Transformer with the gate on its pooled features), two permuted datasets and one split dataset, $T \leq 20$, single insertion point for the gate, $n \leq 20$ per cell (R4's pooled cells reach n=40). Larger Transformers, and Transformers with explicit routing modules, remain untested (the audit's pin sentinel would be the cheap first probe). The R2T flip means our transfer claim is an n=20 null, not a mechanistic impossibility proof; cuDNN nondeterminism on convolution bounds achievable reproducibility (measured: same-seed reruns differ by up to 6.8 learning-term points on ResNet), which is why R2T's gates contain no bit-identity clauses — while R4's Transformer driver proved fully deterministic on its stack, letting its calibration-consistency gate run bitwise. R3's dose curve is six points on a log axis; we did not densify because the preregistered densification trigger (restoration) did not fire, and R4's λ sweep is likewise six decades at one depth. DeLP itself remains non-competitive with replay baselines (Anonymous, 2026b;a); nothing here changes that, and our subject is the structure of the repair's degrees of freedom, not its leaderboard position.

11 Conclusion

One degree of freedom — anchor-budget direction — moves the gain–retention frontier of an active-dendrite repair on the trunk where its supply exists, with clean direction specificity and specificity controls. Four other escapes fail in instructing ways: recalibration restores a shadow of the effect on the toy conv and nothing on the real backbone or the Transformer — where, however, the normalization itself transfers at 40/40 seeds, showing the two freedoms the MLP conflates are mechanically independent; the context pathway cannot

Table 4: The eight implementation-gate failures. "Resolution" is what was done to the *data or code*, never to a criterion; every preregistration file itself remained byte-identical throughout. Gates are deterministic checks in the reproduction package, so each failure is replayable.

#	exp.	gate that fired	root cause → resolution
1	R1	positivity clause mathematically unsatisfiable	prereg also specified Bernoulli sampling, but $P(\text{zero mislabels})=(1-\rho)^T \neq 0$ conflicts with a per-run positivity requirement → resample with a guaranteed-miss rule; all 280 runs rerun (untouched runs verified bit-identical)
2	R2	historical alignment	reference data produced in a CPU era compared against GPU runs through a discrete top-k operator → regenerate references on the same device; threshold unchanged
3	R2	equivalence evidence void	the α=0 equivalence proof lived in the pilot that gate 2 voided → independent 80-run sentinel grid, 40/40 pairs bit-identical
4	R2T	completeness check	checker read a schema-incompatible field name (`avg` vs. `final_avg`) → fix the checker; data untouched; zero analyses had leaked
5	R3	pilot crash	conv anchor constructor had never executed on GPU (all historical conv runs were CPU), exposing a latent device bug → runtime patch in the sweep script only; upstream files unmodified
6	R2T	sample-extension trigger	fired on schedule at n=10 and *overturned* the round's positive conclusion → no resolution needed: the system working as designed
7	R4	weight conservation at λ=0	weights were computed from the anchor count, but a λ=0 run builds no anchors — empty weight vector fails conservation on the first probe run → weights recomputed as a function of the phase index (bit-identical for every $\lambda > 0$); FAIL log retained
8	R4	checker pathway mismatch	prereg required EPOCHS/BS/NTRAIN "written and logged"; the driver inherits them by import (the anti-drift device itself), so the checker's source-text regex matched nothing and aborted before any statistics → checker asserts the imported module constants; criterion values unchanged

carry the signal in training any more than at evaluation; and the discipline that produced these answers caught eight implementation failures, one of which overturned a positive result we would otherwise have reported. The repair family is now mapped across five trunks: its reach is set by the trunk it sits on, and the map is the contribution.

Reproducibility

All experiments are scriptable from the reproduction package (`r_repro/`): preregistrations, revision notes, raw run records, and an independent verifier that recomputes every headline number in this paper from raw

JSON (`python r_repro/verify_package.py`, 25/25 checks; verified on two machines). The 1,539 newly collected runs decompose as 1,506 authoritative — R1 grid 280 plus its dose probe 140; R2 grid 240, α=0 sentinel 80, GPU-reference 80, anti arm 40; R2T 30+30+20; R3 grid 240; R4 gate probe 6, λ-sweep 120, main grid 200 — plus 33 pilot runs; the 280 runs of R1's voided first grid are excluded from every count. A second, paper-level checker (`python check_paper_numbers.py`, 27/27 checks) re-derives the seed-direction counts, per-dataset splits, figure-caption numbers, and every protocol-table constant from the same raw JSON and asserts their agreement with the text — including a regression rule raised from this project's ninth gate failure: caption-level numbers are verified by script, not by eye. R4 contributes its driver, calibration-sweep and main-grid JSONs, its analyzer, and its preregistration/revision/report documents under `results_r4/`. Gate arithmetic, Holm families, and trigger clauses are as fixed in the preregistrations; deviations are logged, preregistrations unmodified.

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